

The cyclic-group hypercontractivity problem is now complete

Literature answer to the open scope in arXiv:2512.03489

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Status. `literature_already_answered`. The cited papers give a full characterization for every cyclic order. Yao’s May 2026 continuation proves the sharp result for every $n \geq 4$, and Xie–Zhang independently prove the same sharp log–Sobolev inequality. A February 2026 paper of Cao–Fan–Han–Qiu–Wang gives the different exact answer for the exceptional group $\mathbb{Z}_3$. Together with the classical two-point case, these papers determine the optimal time for every $n \geq 2$.

Question in the source

For $\psi_n(k) = \min(k, n - k)$, let the Poisson-like semigroup on $\mathbb{Z}_n$ be

$$P_t \left(\sum_{k=0}^{n-1} a_k \chi_k \right) = \sum_{k=0}^{n-1} e^{-t\psi_n(k)} a_k \chi_k.$$

For $1 < p \leq q < \infty$, the problem is to determine the least $t_{p,q}(n)$ such that

$$\|P_t f\|_q \leq \|f\|_p \quad (t \geq t_{p,q}(n)). \quad (1)$$

The source [1] proves

$$t_{p,q}(n) = \frac{1}{2} \log \frac{q-1}{p-1}$$

for $n = 2^k$ and $n = 3 \cdot 2^k$, $k \geq 1$, via the sharp log–Sobolev inequality

$$\text{Ent}_{\mu_n}(f^2) \leq 2 \langle f, A_{\psi_n} f \rangle. \quad (2)$$

Its abstract then says that the general case for arbitrary n remains open.

Explicit completion for every $n \geq 4$

Yao’s continuation [2, Theorems 1.1 and 1.2] states, for every integer $n \geq 4$,

$$\boxed{\|P_t f\|_q \leq \|f\|_p \iff t \geq \frac{1}{2} \log \frac{q-1}{p-1}} \quad (1 < p \leq q < \infty), \quad (3)$$

and proves (2) with the optimal constant 2. This is explicitly presented as the complete solution and cites arXiv:2512.03489 as the prior dyadic-tower result.

The proof does not merely add more towers. It reduces the n -LSI to the nonnegativity of an explicit polynomial of degree at most three in $\lfloor n/2 \rfloor$ variables, then treats all odd $n \geq 5$ and all even $n \geq 4$. Thus every cyclic order omitted by the source is covered.

There is also an independent confirmation. Xie and Zhang [3, Theorem 1 and Corollary 2] prove for every $n \geq 4$ the same sharp inequality (2), and consequently exactly (3), using a different cubic-majorant/Fourier-block method. Their introduction explicitly says that Yao obtained the result first and that their proof was carried out independently.

The exceptional order $n = 3$

Order three is genuinely different: the threshold in (3) is generally false. To state the exact answer, write T_r for the multiplier $r^{\psi_3(k)}$. Since $P_t = T_{e^{-t}}$, if $r_{p,q}(\mathbb{Z}_3)$ is the largest r for which $\|T_r f\|_q \leq \|f\|_p$, then

$$t_{p,q}(3) = -\log r_{p,q}(\mathbb{Z}_3). \quad (4)$$

Cao–Fan–Han–Qiu–Wang [4, Theorem 1.2] prove that, for $1 < p < q < \infty$,

$$\boxed{r_{p,q}(\mathbb{Z}_3) = \frac{(1+2x)(1-y)}{(1+2y)(1-x)}}, \quad (5)$$

where (x, y) is the unique solution in $(0, 1)^2$ of

$$\frac{1}{1+2x} \left(\frac{1+2x^p}{3} \right)^{1/p} = \frac{1}{1+2y} \left(\frac{1+2y^q}{3} \right)^{1/q}, \quad (6)$$

$$\frac{(1-x)(1-x^{p-1})}{1+2x^p} = \frac{(1-y)(1-y^{q-1})}{1+2y^q}. \quad (7)$$

Equations (4)–(7) are an exact finite system and simultaneously identify a nonconstant extremizer. When $p = q$, of course $t_{p,p}(3) = 0$.

Full classification and scope

Combining the cited results gives

$$\boxed{t_{p,q}(n) = \begin{cases} \frac{1}{2} \log \frac{q-1}{p-1}, & n = 2 \text{ or } n \geq 4, \\ -\log r_{p,q}(\mathbb{Z}_3), & n = 3, \end{cases}} \quad 1 < p < q < \infty, \quad (8)$$

with $r_{p,q}(\mathbb{Z}_3)$ given uniquely by (5)–(7). This fully closes the source’s arbitrary-order hypercontractivity problem. Formula (3) also gives the full constant-2 LSI answer requested in the source for all $n \geq 4$; $n = 3$ is exceptional rather than a missing instance of that claim.

The cheap run indexes were searched for all four arXiv identifiers, titles, cyclic hypercontractivity, and word-length LSI phrases; no prior packet was found. Current web searches on 12 August 2026 located the explicit completion, the independent proof, and the exceptional-order characterization.

References

- [1] G. Yao, *Optimal Hypercontractivity and Log-Sobolev inequalities on Cyclic Groups $\mathbb{Z}_{m \cdot 2^k}$* , arXiv:2512.03489 (2025).
- [2] G. Yao, *A Complete Solution of Optimal Hypercontractivity and Log-Sobolev inequalities on Cyclic Groups $\mathbb{Z}_n$ for $n \geq 4$* , arXiv:2605.30867 (2026).
- [3] X. Xie and H. Zhang, *Sharp Log-Sobolev Inequalities for Finite Cyclic Groups with Word-Length*, arXiv:2606.02847 (2026).
- [4] J. Cao, S. Fan, Y. Han, Y. Qiu, and Z. Wang, *The Optimal Hypercontractive Constants for $\mathbb{Z}_3$ and Biased Bernoulli Random Variables*, arXiv:2602.17248 (2026).